

Knowledge-Cryptography Trapdoor Extraction over Native QLever Indexes

Operational Specification (Spec-Proposed; Empirical Tier Labeled **TARGET**—Unmeasured)

ggen / kc-trapdoor work order

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1 Scope and Epistemic Status

This spec is grounded in exactly one real artifact: `/Users/sac/qllever-fortune5-test`, index `fortune5`, builder `git-hash 1075455`, **203 triples**, 47 subjects, 140 objects, 22+7 predicates, all six index permutations on disk (`.spo/.pso/.pos/.osp/.sop` and internal `.pso/.pos` variants), built 2026-08-26. Everything else—buffer sizes, timeouts, latency—is spec-proposed and labeled **TARGET** (unmeasured). No benchmark number in this document was produced by execution against this corpus. The billion-triple regime is future work; nothing here claims it.

2 The Needle-vs-Haystack Invariant

2.1 Statement

Let G be the realization graph (SBB = single-binding-bound, ABB = all-bindings-bound realizations of the SHACL surface). The invariant: *realization of a needle must be parameterized by exact cryptographic coordinates, never by open pattern matching*. Concretely, a needle is addressed only by the pair

$$(H_s = \text{BLAKE3}_K(\text{canonical}(s)), P_\sigma = \text{BLAKE3}_K(\text{salted}(p)))$$

where K is a key held outside the query plane and the hash is computed off-band. Any query whose WHERE clause leaves a transitive-reachability variable un-anchored (free *src/dst* of a ⁺ path) violates the invariant and is refused at the admission gate (Section 4).

2.2 CONSTRUCT via merge joins over pre-sorted arrays

QLever’s six permutations (`fortune5.index.spo`, `.pso`, `.pos`, `.osp`, `.sop`) are pre-sorted ID arrays. A keyed coordinate lookup is a binary search: $O(\log N)$ per bound term, then a merge join across the two operand permutations. The operational CONSTRUCT is `queries/needle-realization.construct.rq`: it binds the needle and salt IRIs from off-band BLAKE3 literals, resolves `kc:salt` $\rightarrow$ `sh:path` against the real index relations (`f5:serviceId`, `f5:region`, `f5:maxReplicationLagMs`, `f5:lastRotatedAt`, `f5:maxAgeDays`, `f5:P95Latency`, `f5:P99Latency`, `f5:availabilitySlaTarget`, `f5:rotateCredential`), then joins the SHACL shape surface. The **<50ms end-to-end latency is a TARGET**: it has not been measured on this corpus.

3 Complexity-Asymmetry Argument

Keyed retrieval is binary search over a sorted permutation: $O(\log N)$, cheap, per-coordinate. The unkeyed alternative—recovering a needle by subgraph-isomorphism search over the open graph—is NP-complete in general [1], and the practical adversary cost is dominated by join-order search over un-anchored transitive closures. The asymmetry is the security property: holders of K pay $\log N$; non-holders pay isomorphism. That is the trapdoor.

References

- [1] M. R. Garey and D. S. Johnson, *Computers and Intractability*. Freeman, 1979.

4 Algorithmic Tar Pits and Fuel Budgets

4.1 Saturated cyclic cliques ($b > 100$)

A trap predicate (here `f5:rotateCredential`, present in the index in `sh:path` position) can be laid out cyclically so that the b -clique closure is saturated. Fuel rule: clique enumeration runs only for a keyed root (`queries/tarpit-clique-guard.select.rq`) with `HAVING(... > 100)` as the saturation detector; the un-anchored closure over the same predicate is refused.

4.2 Recursive path traps

The only permitted form of a $^+$ traversal is *root-anchored with explicit LIMIT* (`queries/tarpit-recursive-path-tr LIMIT 1000`). The refused form is recorded in `queries/unanchored-traversal-refusal.select.rq`: a free-variable (`f5 : connects`) $^+$ enumerates the full closure before any join reduction; it must be rejected at the admission gate before the engine sees it.

4.3 QLever configuration parameters

Buffer sizes, cache caps, and timeouts in `qllever/Qleverfile.fortune5` are spec-proposed; the only measured facts it carries forward from the real Qleverfile are `NAME`, `INPUT_FILES`, `PORT 7020`, `ACCESS_TOKEN`, and `SETTINGS_JSON` (with `ascii-prefixes-only` changed; the change is flagged in-line). Fuel enforcement is two-layer: (1) admission gate refuses un-anchored closures with `KC_FUEL_UNANCHOR 0`; (2) anchored traversals carry an explicit `LIMIT 1000` and a server timeout `KC_TIMEOUT_MS = 50`, both `TARGET`.

5 Zero-Oxigraph Guarantee

The execution plane is native QLever index artifacts only: reads go to `fortune5.index.*` and `fortune5.internal.index.*` via the QLever server for index `fortune5` (builder 1075455). No Oxigraph library, binary, or embedded store appears in the query path; no RDFlib fallback execution is permitted for these queries. Any ticket that routes a KC-trapdoor query through a non-QLever engine is out of scope and refused.

6 Falsifiers

- A measured $p50 > 50\text{ms}$ for the anchored CONSTRUCT on this corpus falsifies the TARGET label (and would need re-scoping).
- An un-anchored $(f5 : \textit{connects})^+$ query admitted for execution falsifies the fuel-budget invariant (Section 4).
- Any query artifact executing against a non-native store falsifies the zero-Oxigraph guarantee (Section 5).